

DESIGN DOCUMENT

Proposed Hybrid Extended Hamming (12,7) Encoder

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Code Architecture

The design adds P0 to Hamming (11,7) and evaluates two restoration placements.

$$P0 = D1 \text{ XOR } \dots \text{ XOR } D7 \text{ XOR } P1 \text{ XOR } P2 \text{ XOR } P4 \text{ XOR } P8 \quad (1)$$



Hybrid-B: restoration is applied only to externally observed parity outputs.

Figure 1. Original Hybrid-B architecture.

Circuit-Level Alternatives

CMOS uses 12-transistor XOR cells. The reconstructed FS-GDI cell uses six transistors. Hybrid-A adds local CMOS restoration to every cell. Hybrid-B adds non-inverting restoration only at parity outputs.

Measured Trade-off

Architecture	Power (uW)	Delay (ps)	PDP (fJ)	Swing (V)	Transistors	Status
CMOS	27.0237	344.947	9.3217	1.2064	264	Pass
FSGDI	2.5502	429.062	1.0942	1.2135	132	Pass
HYBRID_A	25.8367	278.783	7.2029	1.2039	220	Pass
HYBRID_B	11.2029	215.846	2.4181	1.2042	152	Pass

Hybrid-B is selected for speed; FS-GDI remains the energy optimum under the tested nominal condition.